

PostgreSQL 18 Separate Server Setup

CMD/PowerShell se server create/start karna, phir pgAdmin 4 se manage karna

Final target setup	PostgreSQL 18 normal server -> localhost:5432 ecommerce-postgres-18 server -> localhost:5434
Use case	Kotlin/Spring Boot microservices ke liye separate PostgreSQL server and multiple DBs
pgAdmin role	Server register karna, DB create karna, tables/query manage karna
CMD role	New PostgreSQL instance initialize/start karna and port verify karna

Important note: pgAdmin 4 se new PostgreSQL server/instance create nahi hota. pgAdmin sirf already running server ko connect/manage karta hai.

1. Why CMD/PowerShell and why not only pgAdmin 4?

Simple answer: pgAdmin 4 ek GUI client hai. PostgreSQL server actual database engine hai. Agar new server localhost:5434 pe running hi nahi hai, to pgAdmin sirf timeout dega.

Task	CMD/PowerShell	pgAdmin 4
New PostgreSQL data folder create karna	Yes	No
initdb run karke new instance initialize karna	Yes	No
postgresql.conf me port set karna	Yes	No
Server process start/stop karna	Yes	No
Server running hai ya nahi check karna	Yes	No
Register Server connection entry banana	No	Yes
Database/table/query manage karna	Possible, but not convenient	Yes

Real example

Tumhare laptop me already ek normal PostgreSQL 18 server hai on localhost:5432. Tum ecommerce project ke liye second server chahte the on a different port. Isliye new instance create/start karna zaroori tha.

```
PostgreSQL 18 normal server -> localhost:5432
ecommerce-postgres-18 server -> localhost:5434
```

2. What exactly we did in CMD/PowerShell

Step 1 - New data folder create kiya

Yeh folder ecommerce PostgreSQL server ke data files ke liye hai.

```
mkdir D:\PostgreSQL\ecommerce-18-data
```

Step 2 - New PostgreSQL instance initialize kiya

initdb command ne new PostgreSQL cluster create kiya. Isse postgresql.conf, pg_hba.conf, base, global, WAL files etc. generate hue.

```
& "C:\Program Files\PostgreSQL\18\bin\initdb.exe" -D "D:\PostgreSQL\ecommerce-18-data" -U postgres -W -E UTF8
```

Is command me password enter kiya tha. pgAdmin me connect karte time wahi password use karna hai.

Step 3 - Port active kiya

Config file open karke port line ko active kiya. Pehle line commented thi:

```
#port = 5434
```

Correct active line:

```
port = 5434
```

Important: # ka matlab comment hota hai. Agar # laga rahega to PostgreSQL us line ko ignore karega.

Step 4 - Server start kiya

pg_ctl command se ecommerce PostgreSQL server start hua.

```
& "C:\Program Files\PostgreSQL\18\bin\pg_ctl.exe" -D "D:\PostgreSQL\ecommerce-18-data" -l "D:\PostgreSQL\ecommerce-18.log" start
```

Step 5 - Server running hai ya nahi verify kiya

netstat se check kiya ki 5434 pe PostgreSQL listen kar raha hai ya nahi.

```
netstat -ano | findstr ":5434 "
```

Successful output example:

```
TCP 127.0.0.1:5434 0.0.0.0:0 LISTENING 30152  
TCP [::1]:5434 [::]:0 LISTENING 30152
```

LISTENING ka matlab server start ho gaya and connection accept kar sakta hai.

3. What we do in pgAdmin 4 after server starts

Server start hone ke baad pgAdmin 4 ka kaam start hota hai. Ab pgAdmin se Register Server, databases, tables, and queries manage kar sakte ho.

Step 1 - Register Server

- 1 pgAdmin 4 open karo.
- 2 Left side me Servers pe right-click karo.
- 3 Register -> Server select karo.
- 4 General tab me name do: ecommerce-postgres-18.
- 5 Connection tab me details fill karo.

Field	Value
Host name/address	127.0.0.1 or localhost
Port	5434
Maintenance database	postgres
Username	postgres
Password	initdb me jo password dala tha

Step 2 - Databases create karo

Connect hone ke baad ecommerce-postgres-18 server ke andar Databases pe right-click -> Create -> Database.

- user_db
- product_db
- order_db
- inventory_db
- notification_db

Ya Query Tool se SQL run kar sakte ho:

```
CREATE DATABASE user_db;  
CREATE DATABASE product_db;  
CREATE DATABASE order_db;  
CREATE DATABASE inventory_db;  
CREATE DATABASE notification_db;
```

4. Spring Boot application.yml configuration

Ab tumhare Kotlin/Spring Boot microservices ko ecommerce PostgreSQL server se connect karna hai. Iske liye port 5434 use karo.

user-service

```
spring:
  datasource:
    url: jdbc:postgresql://localhost:5434/user_db
    username: postgres
    password: your_password
```

product-service

```
spring:
  datasource:
    url: jdbc:postgresql://localhost:5434/product_db
    username: postgres
    password: your_password
```

order-service

```
spring:
  datasource:
    url: jdbc:postgresql://localhost:5434/order_db
    username: postgres
    password: your_password
```

PostgreSQL me normally username postgres hota hai. MySQL jaisa root use karoge to password authentication failed error aa sakta hai.

5. Daily use commands

Laptop restart ke baad manually created server stop ho sakta hai. Dobara start karne ke liye ye command use karo.

Start server

```
& "C:\Program Files\PostgreSQL\18\bin\pg_ctl.exe" -D "D:\PostgreSQL\ecommerce-18-data" -l "D:\PostgreSQL\ecommerce-18.log" start
```

Stop server

```
& "C:\Program Files\PostgreSQL\18\bin\pg_ctl.exe" -D "D:\PostgreSQL\ecommerce-18-data" stop
```

Restart server

```
& "C:\Program Files\PostgreSQL\18\bin\pg_ctl.exe" -D "D:\PostgreSQL\ecommerce-18-data" restart
```

Check port

```
netstat -ano | findstr ":5434 "
```

Check latest log

```
Get-Content "D:\PostgreSQL\ecommerce-18.log" -Tail 80
```

6. Common problems and fixes

Problem	Reason	Fix
pgAdmin timeout on 5434	Server is not running on that port	Run <code>pg_ctl start</code> and check <code>netstat LISTENING</code>
<code>#port = 5434</code>	Line is commented, PostgreSQL ignores it	Remove <code>#</code> so it becomes <code>port = 5434</code>
password authentication failed for user root	PostgreSQL user root is wrong or password incorrect	Use username postgres and correct password
<code>findstr :5433</code> shows 54336	54336 contains 5433 text but is not port 5433	Use <code>findstr ":5434 "</code> with space for exact match
Could not create TCP/IP sockets	Port/bind permission issue	Try another port like 5434/5440 and run PowerShell as Administrator
pgAdmin Register Server fails	Register Server does not create a server	Start actual PostgreSQL instance first, then register it

7. Interview/explanation line

Agar koi puche why CMD and pgAdmin both, you can explain:

pgAdmin 4 is only a GUI client to manage an already running PostgreSQL server. I wanted a separate PostgreSQL 18 instance for my ecommerce microservices project on a different port, so I initialized a new data directory, configured the port, and started the server using command line. After the server was running, I used pgAdmin 4 to register the server and manage databases.

8. Final summary

- CMD/PowerShell ka kaam: create instance, set port, start server, verify port.
- pgAdmin 4 ka kaam: connect/register server, create databases, create tables, run queries.
- Your ecommerce server is localhost:5434.
- Spring Boot microservices should use `jdbc:postgresql://localhost:5434/`.